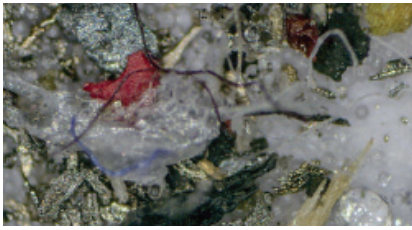


Addressing particle contamination on an electrical components/PCBs



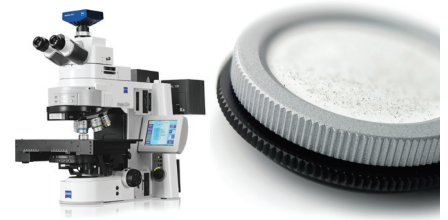
Technical cleanliness is a two-stage procedure that uses a light microscope to examine the number and size of particles present, followed by an electron microscope to confirm the chemical composition and origin of the particles.

The particles may also differ in character – while combustion engines typically encounter hard and abrasive metallic particles, electronic engines can be susceptible to low- and high-conductivity particles. Most particles are

generated during the processing of components and assemblies (approx. 80%) and only a fraction (approx. 20%) are due to environmental influences.

When it comes to PCBs, any metallic particles that measure $> 200 \mu\text{m}$ or more present a potential risk of electrical short circuit. If the metallic particles are of a certain length, angle and position, and they land on the PCB in a particular way, they can cause a short circuit. This is what makes it important to find these critical particles and detach them from the PCBs..

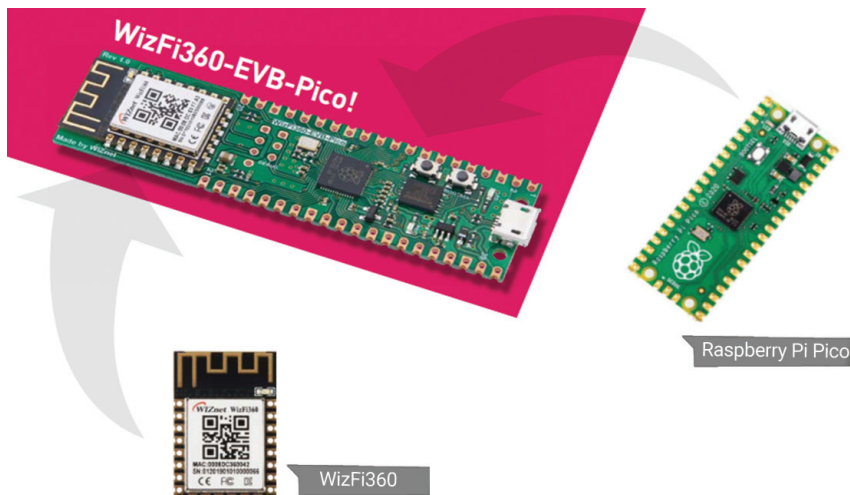
Particle contamination has been a growing problem for the automotive industry since the early 1990s, when systems started becoming increasingly complex and installation spaces ever



smaller. VDA 19/ISO 16232 was therefore established as a standard for enabling customers and suppliers to address the risk of potential damage of products across the production chain. In the electronics industry, VDA 19 does not specify any limiting values for component cleanliness. These must be defined according to component function, producibility, and verifiability.

To know more on Technical Cleanliness in PCB Inspection mail us at info.metrology.in@zeiss.com or call on +91 080 4343 8222

Meet our latest board WizFi360-Evb-Pico!



WIZnet will be holding a design contest in June and offer free samples to our participants to encourage the advancement of IoT ecosystem and to coexist with other vendors. While the focus of the last contest was ethernet functionality, this upcoming contest will primarily focus on Wi-Fi functionality featuring our WizFi360 products. WizFi360 is registered as Arm's official shield for Open-CMSIS-Pack and is officially registered and used on various platforms such as Azure Certified. WizFi360 supports AWS AT Command as well as AWS SDK, and also supports various development environments, such as Arduino, C/C++, Python, Free-RTOS, and more.

In May 2022 WIZnet released WizFi360-EVB-Pico - RP2040-based evaluation board for WizFi360 Wi-Fi Module. It is pin compatible with Raspberry Pi Pico board and can be used for IoT Solution development. Available for purchase at \$6.95 WizFi360-EVB-Pico is one of the most cost-effective and powerful solutions for prototyping home and industrial IoT devices.

To find more details, visit:

<https://docs.wiznet.io/Product/Open-Source-Hardware/wizfi360-evb-pico>

SDK • C/C++ • MicroPython • CircuitPython

More details at:
maker.wiznet.io

Contact:
muralidharan@wiznet.io

